

Assignment

In this homework, I deployed and tested a local Large Language Model (LLM) using Ollama, Python, Jupyter Notebook, Docker, and Open WebUI. The goal was to demonstrate text generation, multimodal capabilities, and containerized execution.

1. Writing a Python Prompt to the LLM

Code

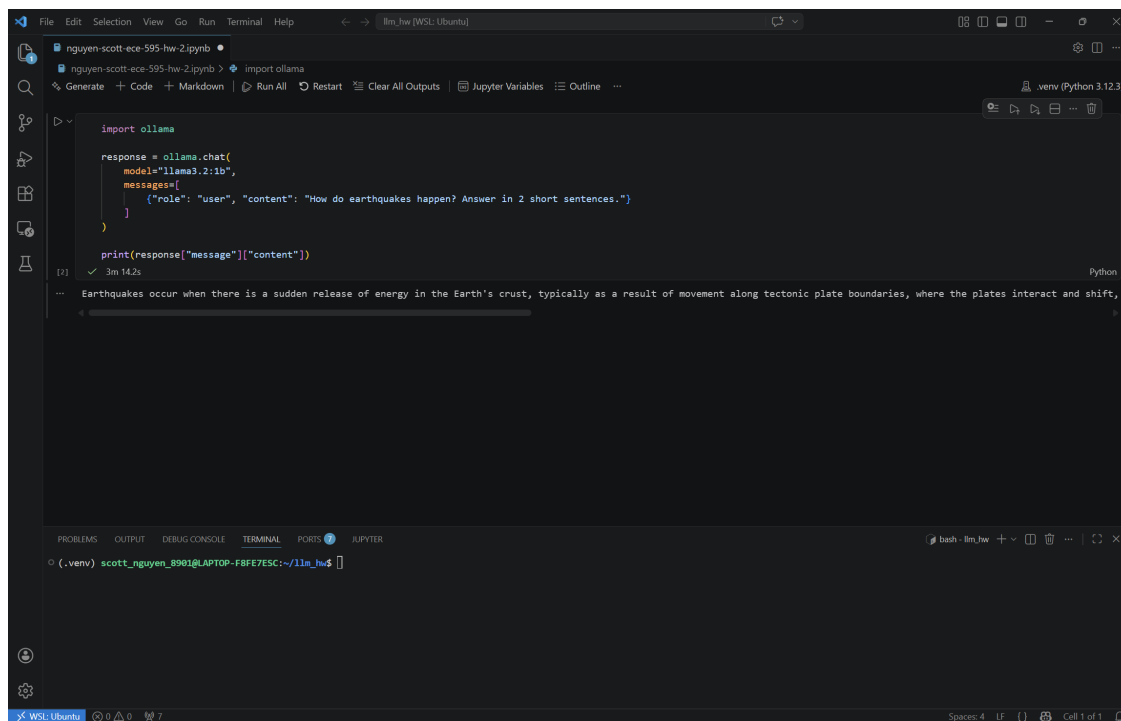
```
import ollama

response = ollama.chat(
    model="llama3.2:1b",
    messages=[
        {"role": "user", "content": "How do earthquakes happen? Answer in 2 short sentences."}
    ]
)

print(response["message"]["content"])
```

Output

```
Earthquakes occur when there is a sudden release of energy in the Earth's crust, typically as a result of movement along tectonic plate boundaries, where the plates interact and shift, causing stress to build up and eventually lead to a rupture or tremor. This process is known as seismogenesis, and it can be triggered by changes in pressure, temperature, or other factors that cause the rocks to fracture and release energy.
```



The screenshot shows a Jupyter Notebook interface with a dark theme. The notebook is titled 'nguyen-scott-ece-595-hw-2.ipynb'. The code in the cell is as follows:

```
import ollama

response = ollama.chat(
    model="llama3.2:1b",
    messages=[
        {"role": "user", "content": "How do earthquakes happen? Answer in 2 short sentences."}
    ]
)

print(response["message"]["content"])
```

The output of the cell is displayed below the code, showing the generated text: "Earthquakes occur when there is a sudden release of energy in the Earth's crust, typically as a result of movement along tectonic plate boundaries, where the plates interact and shift,". The output is truncated with an ellipsis on the left and a right arrow on the right. The bottom of the interface shows a terminal window with the command prompt 'bash - llm_hw' and the user 'scott_nguyen_8981@LAPTOP-FBF7E7ESC:~/llm_hw\$'.

Figure 1: Text generation using Ollama

2. Multimodal Model (Image)

Code

```
import ollama

response = ollama.chat(
    model='llava',
    messages=[
        {
            'role': 'user',
            'content': 'Describe this image in 2 sentences.',
            'images': ['cat.jpg']
        }
    ]
)

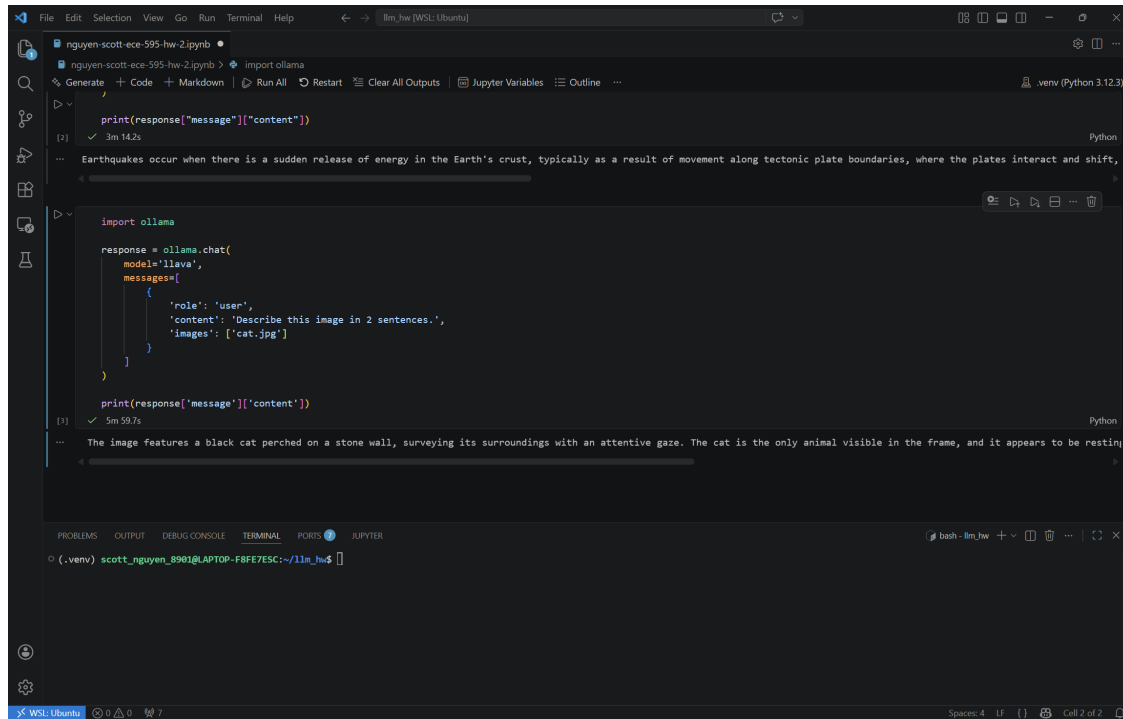
print(response['message']['content'])
```

Output

The image features a black cat perched on a stone wall, surveying its surroundings with an attentive gaze. The cat is the only animal visible in the frame, and it appears to be resting or observing something out of view. The background is blurred but suggests a natural outdoor setting with trees and foliage.



Figure 2: Input image



```
File Edit Selection View Go Run Terminal Help | llm_hw [WSL: Ubuntu]
nguyen-scott-ece-595-hw-2.ipynb •
nguyen-scott-ece-595-hw-2.ipynb > import ollama
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
venv (Python 3.12.3)

print(response["message"]["content"])
[2] ✓ 3m 14.2s Python
Earthquakes occur when there is a sudden release of energy in the Earth's crust, typically as a result of movement along tectonic plate boundaries, where the plates interact and shift,

import ollama
response = ollama.chat(
    model='llava',
    messages=[
        {
            'role': 'user',
            'content': 'Describe this image in 2 sentences.',
            'images': ['cat.jpg']
        }
    ]
)

print(response['message']['content'])
[1] ✓ 5m 59.7s Python
The image features a black cat perched on a stone wall, surveying its surroundings with an attentive gaze. The cat is the only animal visible in the frame, and it appears to be resting

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
(.venv) scott_nguyen_8981@LAPTOP-F8FE7ESC:~/llm_hw$
```

Figure 3: Multimodal output

3. Docker Container

Commands

```
docker build -t llm-hw .  
docker run -p 8888:8888 llm-hw
```

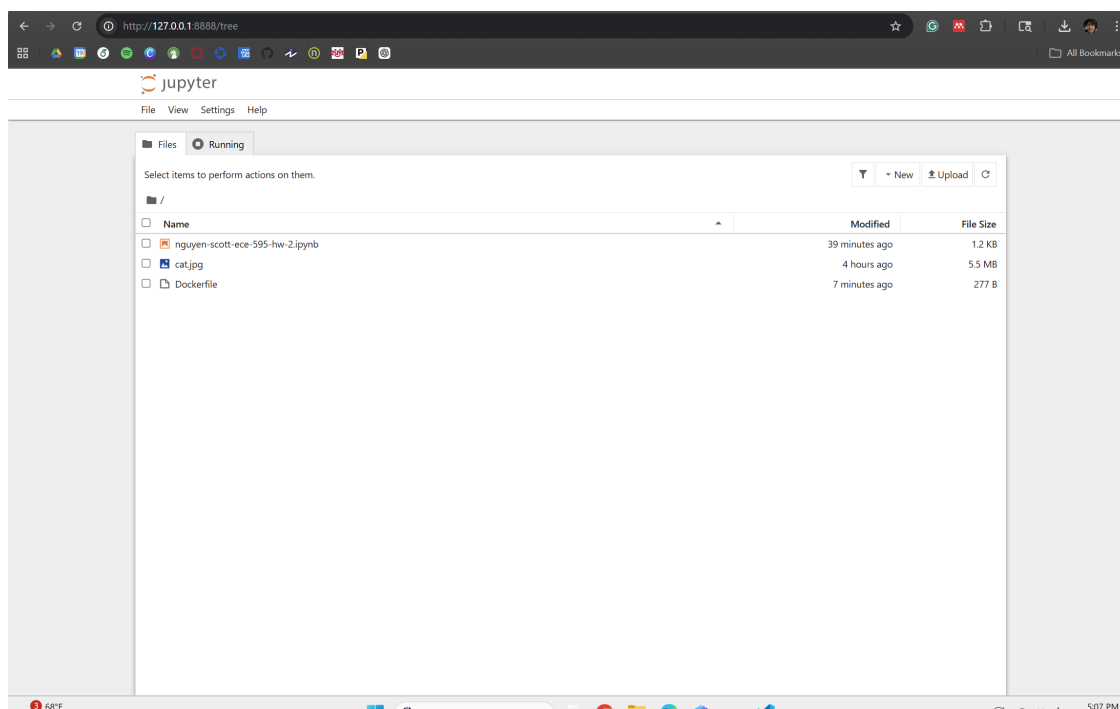
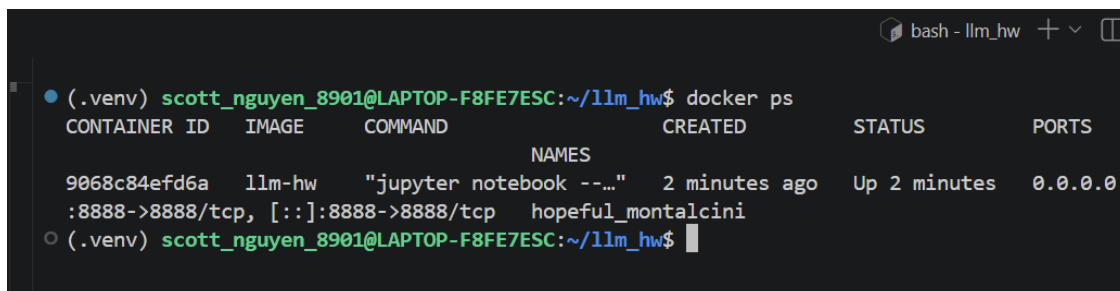


Figure 4: Docker container running

4. Jupyter Notebook in Docker



```
bash - llm_hw + - □
• (.venv) scott_nguyen_8901@LAPTOP-F8FE7ESC:~/llm_hw$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
9068c84efd6a   llm-hw    "jupyter notebook --..." 2 minutes ago  Up 2 minutes  0.0.0.0
:8888->8888/tcp, [::]:8888->8888/tcp  hopeful_montalcini
○ (.venv) scott_nguyen_8901@LAPTOP-F8FE7ESC:~/llm_hw$
```

The screenshot shows a terminal window with the command `docker ps` executed. The output is a table of running containers. One container, `9068c84efd6a`, is running the `llm-hw` image. Its command is `"jupyter notebook --..."`. It was created 2 minutes ago and is currently up for 2 minutes. It is exposed on port 8888, mapping the host's 0.0.0.0:8888 to the container's 8888 on both TCP and UDP. The container name is `hopeful_montalcini`.

Figure 5: Jupyter running in browser

5. Web UI Interaction

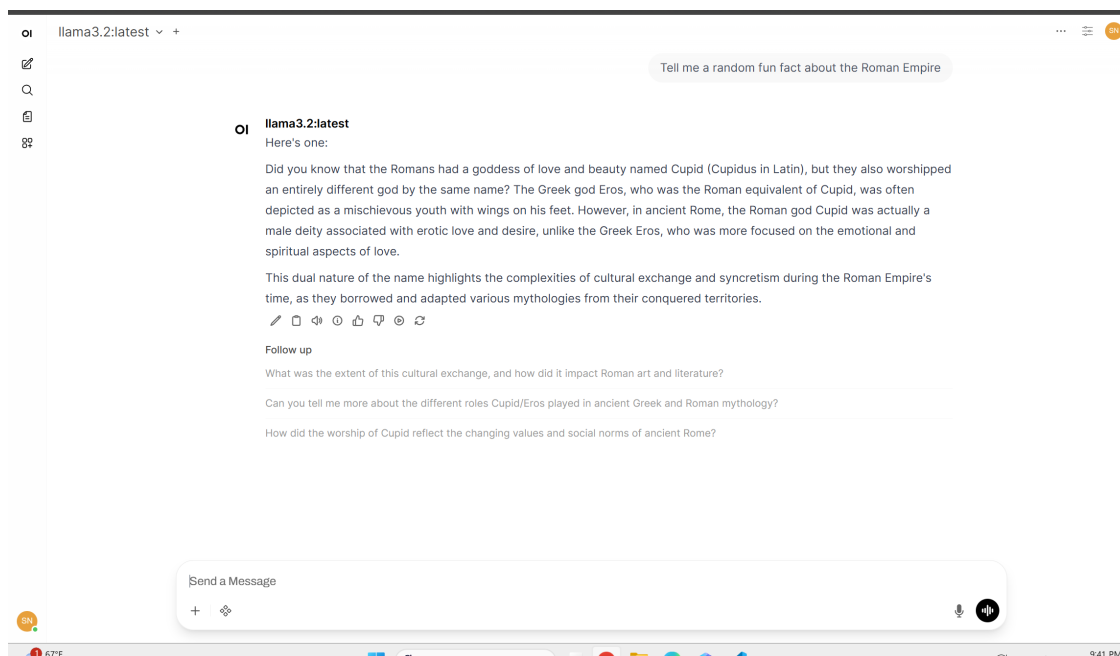


Figure 6: Open WebUI interacting with model

Conclusion

In this assignment, I successfully:

- Ran a local LLM using Ollama
- Generated text responses via Python
- Used a multimodal model for image understanding
- Containerized the application using Docker
- Interacted with the model through a web interface

This demonstrates a complete local AI deployment pipeline.